

1 Part 14: Curse of Dimensionality

2 Summary

3 Often data come to us in a **high-dimensional form**, meaning that each ob-
4 servation is actually best thought of as a vector of many numbers.

5 In some situations, it is crucial that lower-dimensional representations of
6 these data are obtained. Here we present some approaches to doing that.

7 First, we motivate why this dimension reduction can be so important.

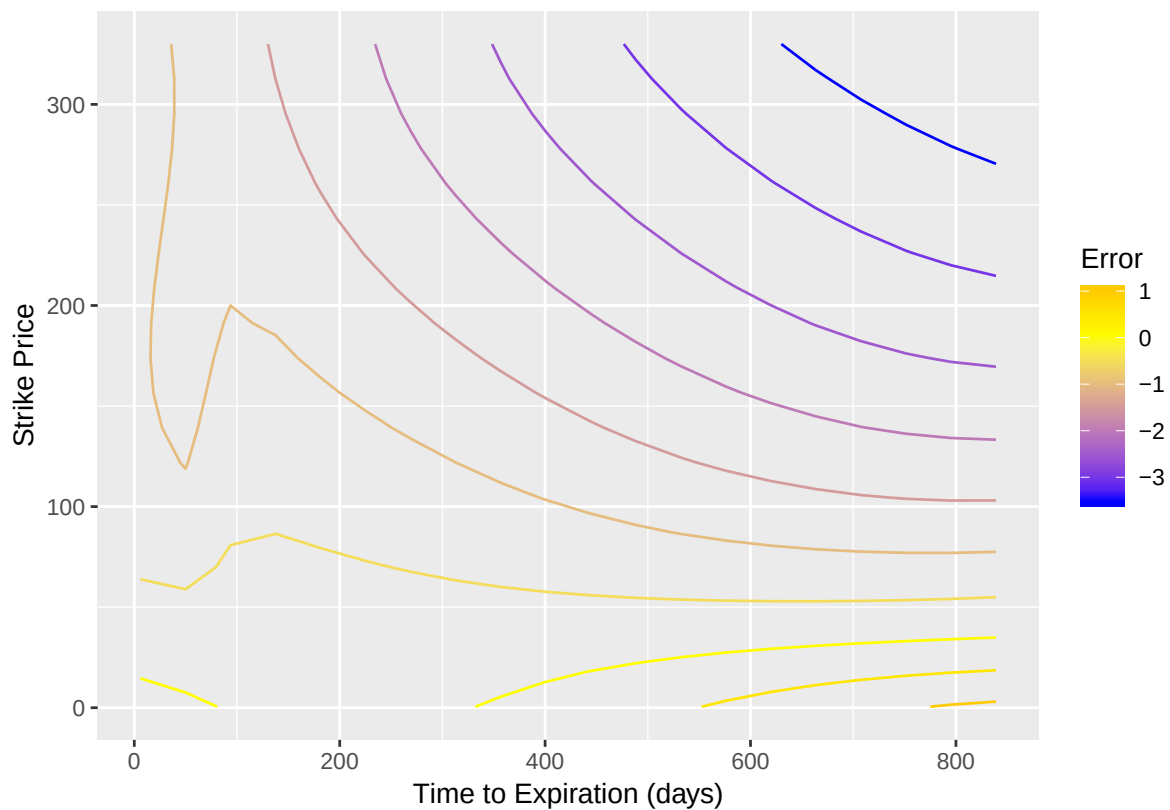
1 The Curse of Dimensionality

2 Despite the promise of nonparametric smoothing approaches, they suffer
3 from a serious drawback: The amount of data required to fit nonparamet-
4 ric regressions well increases exponentially with the dimension of the data.

5 This is called the **Curse of Dimensionality**.

- 1 To start, we ask: **What is the “dimension” of data?**
- 2 Standard data come to us in the form of numbers, but a single **observation**
- 3 is often best thought of as a list of numbers.
- 4 For example, a **yield curve** shows how bond rates vary as a function of
- 5 time to maturity. A single yield curve consists of several numbers, usually
- 6 around ten rates. This count of how many numbers it takes to represent a
- 7 yield curve is called its **dimension**. We would say that a single yield curve
- 8 is “ten-dimensional.”
- 9 Another example: Imagine our data consists of separate price time series
- 10 for each of 100 stocks. Each series goes back 20 trading days. We would say
- 11 in this case that we have “100 observations of 20-dimensional time series”.

- 1 The dimension of data often comes up is in the context of regression or
- 2 supervised learning. If there are p predictors being used in a model for a
- 3 response variable, we would say that we are using a “ p -dimensional pre-
- 4 dictor vector.”
- 5 Nonparametric regression is possible when the predictor vector is p -
- 6 dimensional. For instance, if $p = 2$, a neighborhood can be thought of
- 7 as a square centered on the target value. Local linear regression involves
- 8 fitting a plane within this square.
- 9 Reconsider our options sample from the previous Part. The figure on the
- 10 following page shows a two-dimensional local linear regression fit. The
- 11 response is the “error” in the Black-Scholes price (relative to the ask price)
- 12 and the predictors are the time to expiration and the strike price.



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- 1 **Exercise:** The next plot adds on the 1,000 observed predictor pairs used in
2 this case. Sketch the neighborhood when the regression function is esti-
3 mated at (400, 200). What will have to happen in order for this estimate to
4 have sufficiently low variance?

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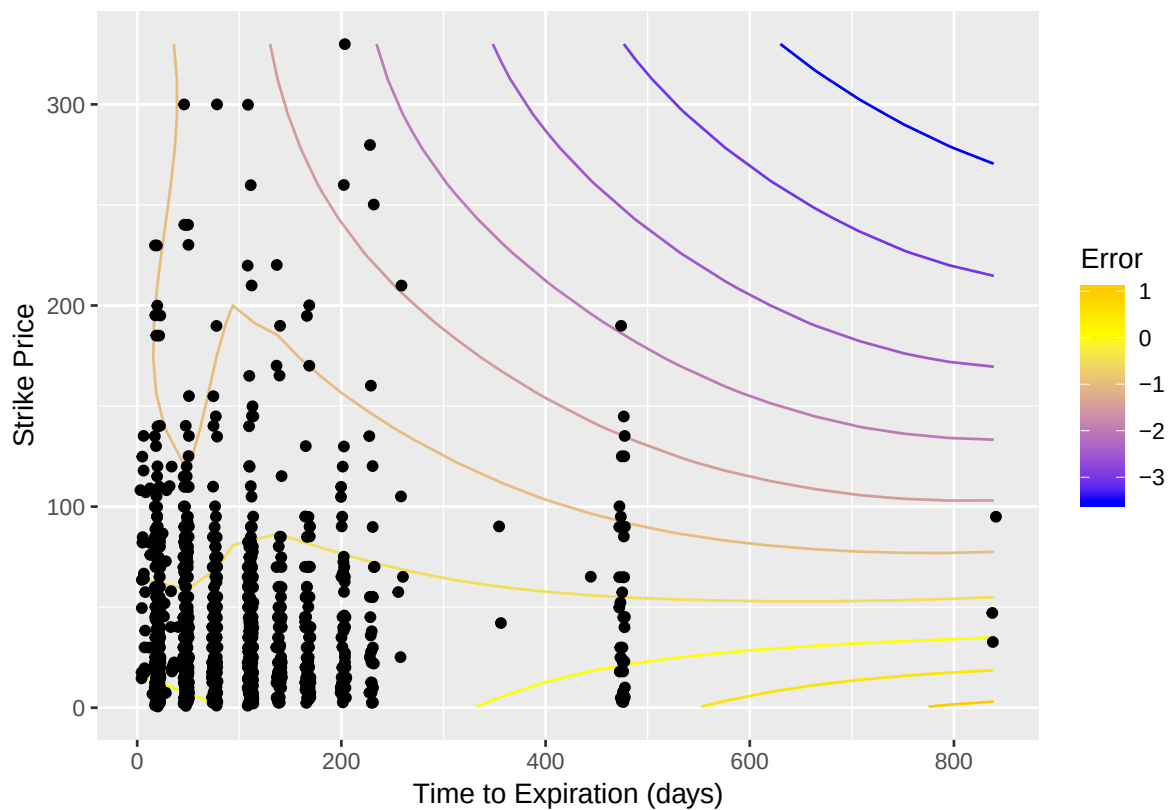
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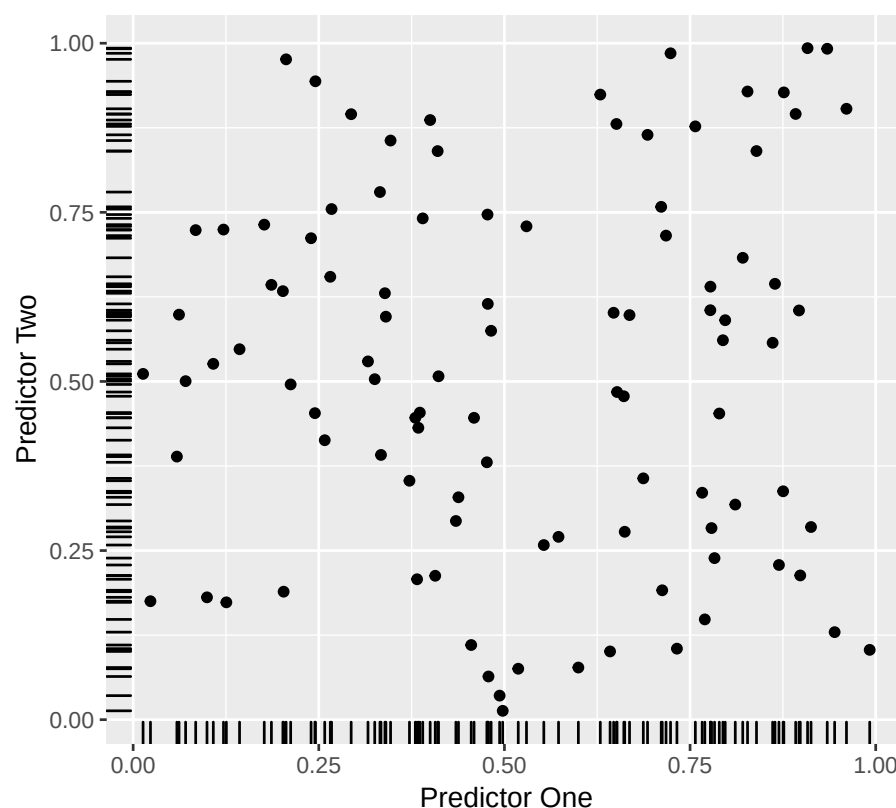
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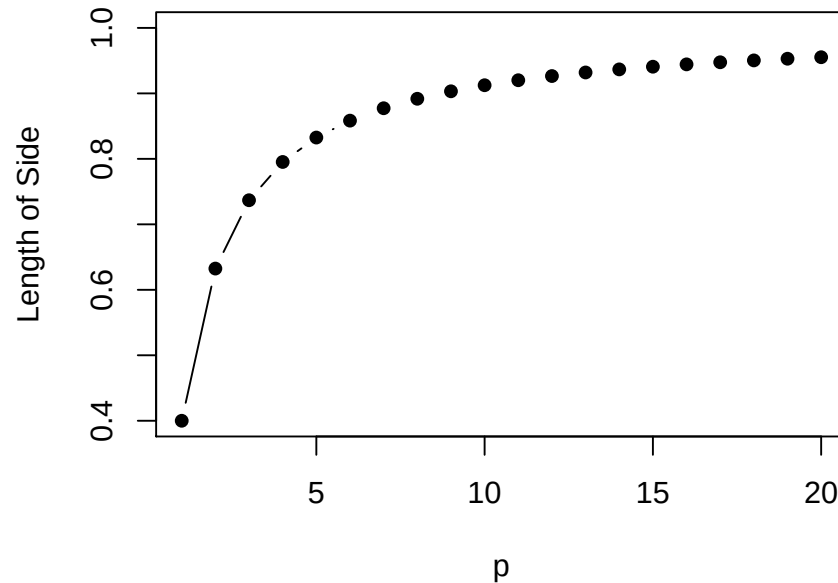
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- 1 This example shows the fundamental challenge of working with high-
- 2 dimensional predictors: As the dimension increases, the available obser-
- 3 vations become more “spread out.” Neighborhoods must be made larger
- 4 in order to compensate and achieve estimates with acceptable variance.
- 5 But, choosing a large neighborhood takes away a key feature of using non-
- 6 parametric approaches: We want the estimate to be flexible, and not be
- 7 based on too much “smoothing.”

- 1 Consider the plot on the following slide.
- 2 Here, the sample size is 100. Each observation is two-dimensional, i.e.,
- 3 imagine these are the two predictors. These data are simulated from uni-
- 4 form distributions.
- 5 The “marks” shown in each margin depict the **marginal distribution** for
- 6 each predictor. Note that in the margins, the data appear to be quite closely
- 7 spaced. If we were to fit a nonparametric regression using just one predic-
- 8 tor, neighborhoods could be chosen small.
- 9 But, in two dimensions, large gaps appear in the distribution of the obser-
- 10 vations. Neighborhoods will have to be chosen larger in order to capture
- 11 enough data.



- 1 Another way of thinking about it: Under the uniform setup, with a p -
2 dimensional predictor, in order for a neighborhood to include proportion
3 α of the data, the length of the sides of the neighborhood must be $\alpha^{1/p}$. This
4 plot shows the case where $\alpha = 0.4$.



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- 1 **Exercise:** Suppose that $p = 5$. Under our uniform setup, how long are
2 the sides of a neighborhood that covers 40% of the data? What are the
3 implications of this?

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1 **Exercise:** Again assume our uniform predictor distribution. Define that an
2 observation is “close to the boundary” if it is either less than 0.1 or greater
3 than 0.9 for at least one predictor. What proportion of the observations are
4 “close to the boundary?”

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1 The curse of dimensionality motivates us to consider methods which cre-
2 ate **low-dimensional representations** of data.

3 In many cases, data which are p -dimensional in their original form can be
4 compressed to q dimensions, where $q \ll p$, with minimal loss of important
5 information.

6 The aforementioned yield curves provide an important example. The daily
7 yield curve consists of approximately 10 rates, but a yield curve takes a
8 limited class of shapes. Shifts in the yield curve can be described using
9 three numbers in most situations. Hence, a 10-dimensional object can be
10 described compressed to three dimensions.